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Author(s): E. G. Pulleyblank

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THE FINAL CONSONANTS OF OLD CHINESE

E. G. PULLEYBLANK

University of British Columbia

For the internal reconstruction of Middle Chinese we have the firm guidelines provided by the rhyme tables and the rhyme dictionaries. For earlier periods such analyses by contemporary Chinese phonologists are lacking and we must depend on what can be inferred indirectly from evidence of various kinds, the most important of which are poetic rhyming and the phonetic elements in the structure of characters.

Chinese poetry made use of rhyme from the earliest times and the fact that the rhyming of the Book of Poetry became increasingly imperfect in the course of time was one of the principal stimuli to the early development of interest in historical phonology in China. It was correctly inferred that the rhymes were imperfect because the sounds had changed and efforts were made with increasing sophistication to determine what the ancient rhyming categories had been. In the absence of a phonetic notation inferences about the actual sounds of the words in ancient times remained vague but the philologists of the Ch'ing period were able to establish an agreed set of rhyming categories which, with small refinements, have remained the basis for studies of the subject to the present day.

The list given below is based (with some rearrangement) on that given in the introduction to the study of Han dynasty rhymes by Lo

Note on Transcriptions

Transcriptions of Chinese characters given in IPA notation are Early Middle Chinese (EMC) unless otherwise specified. By Early Middle Chinese I mean my new reconstruction of the *Ch'ieh-yün* a summary of which appears in slightly outdated form in Pulleyblank 1979. Note that I here indicate rising and departing tones by raised ^ˊ and ^ˋ respectively. In a few cases I give Late Middle Chinese values, marked LMC (see Pulleyblank 1970-71). Reconstructed values earlier than the *Ch'ieh-yün* are marked with an asterisk.

Ch'ang-p'ei and Chou Tsu-mo (1958). Rhymes are classified as *yin*, *yang*, or *ju*, depending on whether their Middle Chinese reflexes ended in (a) a vowel or glide, (b) a nasal, (c) a stop. In cases where *yin* and *yang* rhymes occurred in more than one tone the level tone category is taken to stand for all the tones. Beside each category is given Karlgren's Archaic reconstruction. For each category Karlgren in fact reconstructed several vowels or combinations of vowel and medial but I have chosen in each case the one corresponding to Grade I of the rhyme tables or to 'pure' Grade IV in those rhymes where a Grade I form is lacking. Where two forms are given, it is because Karlgren split a category which other scholars have continued to treat as one (Categories IV(ii)a, VIIIa, Xa), or because he followed a different authority on the rhyme categories from that followed by Lo and Chou (Category V).

	(a) <i>yin</i>	(b) <i>yang</i>	(c) <i>ju</i>
I		侵 əm	緝 əp
II		談 âm	盍 âp
III	微 ər	諄 ən	術 ət
IV(i)	祭 âd	元 ân	月 ât
(ii)	歌 â, âr		
V	脂 ied, ər	真 ien	質 iet
VI	支 ieg	耕 ieng	錫 iek
VII	之 əg	蒸 əng	職 ək
VIII	魚 âg, o	陽 âng	鐸 âk
IX	幽 ôg	冬 ông	毒 ôk
X	侯 ug, u	東 ung	屋 uk
XI	宵 og		藥 ok

It is immediately apparent that there is a very uneven distribution of vowels and final consonants. While the two central vowels, ə and â, occur before all types of finals – labials, dentals and velars – the front vowel, e, occurs only before velars and dentals and the three back vowels, u, ô and o (ɔ), occur only before velars. This seems a very improbable system for a natural language.

In an article published in 1963 I proposed a solution which would generalize the two vowels, ə and â, to all rhyme categories, accounting for the front and back-rounded finals by means of palatal and labial

semivowels placed after or before the nucleus. While the general principle of limiting the distinctions in the nuclear vowel to a two-way opposition based on vowel height was undoubtedly correct, the particular solution proposed was unsatisfactory in a number of ways, not least, because it did not really account for the distributional anomalies. That is, no explanation was provided for the different distribution of semivowels with different types of finals or for the fact that sometimes the rhyme was affected by a semivowel and sometimes it was not. The way out of these difficulties readily emerges, however, if we enlarge the inventory of possible types of final consonants, adding palatals, labiovelars and uvulars. In the development from Old Chinese to Middle Chinese the six original types of consonants were reduced to the three types that still existed at that time but the palatals, labiovelars and uvulars left their traces on the vocalism.

In addition to providing a good solution for the *Shih-ching* rhyme categories this hypothesis ends up with the same inventory of consonant types initially and finally in Old Chinese. It is generally true in the monosyllabic languages of East Asia that they have a simpler set of consonants finally than initially, sometimes drastically so; but it is also generally true that where we can trace the historical development, there has been a strong and continuous trend towards reduced complexity at the end of the syllable. Projecting backward we can hypothesize that there was a stage at which the same consonantal distinctions existed finally and initially. This hypothesis has important implications for the problem of “word families” in proto-Chinese (cf. Pulleyblank 1973a).

Reducing vowel contrasts to two may not at first sight seem natural for a language but there are living languages, namely those of the Northwest Caucasus, which do just that. Moreover, as I have argued elsewhere, there is good reason to think that it has been the underlying pattern in Chinese at all periods. It was recognized in Middle Chinese by the early rhyme table phonologists when they set up their “inner” (ə) “outer” (a) opposition and, with various mutations it can also be recognized in modern Chinese dialects. In Old Chinese it was the basis of an important morphological alternation or ablaut, involving semantic contrast between “introvert” (ə) and “extrovert” (a), e.g. 譚 *dəm* “talk about,” 談 *dam* “talk (intrans.); conversation”; 似 *zi* “resemble,” 象 *ziaŋ* “imitate, image” (Pulleyblank 1965b).

I assume that this alternation was originally available for syllables of all kinds.

General Characteristics of the Old Chinese Rhyme Categories

It is not my intention in this paper to give a complete theory of the evolution of the *Ch'ieh-yün* rhyme system out of the *Shih-ching* categories. It is necessary at the outset, however, to make some remarks about the overall scheme.

(a) Type A and B syllables

One of the key questions in the whole of Chinese historical phonology is the interpretation of Grade III in the rhyme tables, or rather of the great division among *Ch'ieh-yün* rhymes between those that fall wholly or partly into Grade III and those that do not. Karlgren's answer was to reconstruct medial *yod* $\dot{i}$ ($=j$ in IPA notation) in all "Grade III rhymes" (including parts of those rhymes which actually fall into Grades II or IV). In my article on "The consonantal system of Old Chinese" (1962) I rejected the projection of this *yod* back into Old Chinese and proposed to derive it from an earlier length distinction. Since then I have come to the conclusion that Karlgren's *yod* is not valid even for Middle Chinese. In Late Middle Chinese Grades III and IV together were characterized by a vocalic medial $\dot{i}$ in *k'ai-k'ou* and a corresponding front rounded y in *ho-k'ou*, the glide j being found after certain initials in Grades II and IV (Pulleyblank 1970–71). In Early Middle Chinese, that is, the language of the *Ch'ieh-yün* itself, there were three high vocalic medials, $\dot{i}$, $\dot{\epsilon}$, u , which developed into the Late Middle Chinese system by the fronting of $\dot{\epsilon}$ to $\dot{i}$ and u to y (the latter being also partly derived from Early Middle Chinese $w\dot{i}$ and $w\dot{\epsilon}$).

Instead of a contrast between yodized (Grade III) and unyodized (Grades I, II and IV), therefore, we must contrast in Early Middle Chinese syllables with or without a high vowel ($\dot{i}$, $\dot{\epsilon}$, u) immediately after the initial complex. I shall call syllables without such a medial Type A and those with such a medial Type B. When we compare the rhymes of the *Ch'ieh-yün* with the *Shih-ching* categories from which

they emerge, we find that each category gave rise to at least one Type A final and at least one Type B final. When there are more than one of each, one can generally match them in pairs in terms of common features. Thus from the *Shih-ching* *-an* category we have:

	Type A	Type B
plain	-an 寒(ân)	-ian 元(ïen)
<i>j</i> -umlaut	-en 先(ien)	-(j)ian 仙(ïän) Grade IV
<i>r</i> -umlaut	-arn 刪(an)	} -(r)ian 仙(ïän) Grade III
<i>rj</i> -umlaut	-ern 山(ăn)	

(Karlgren's Ancient reconstruction of the *Ch'ieh-yün* rhymes is given in brackets for reference.)

I assume that the split of every *Shih-ching* rhyme category into Type A and B finals reflects a prosodic feature which I symbolize by an acute accent for Type A and a grave accent for Type B: *-ân>-an*, *-àn>-ian*, *·jân>-en*, *·jân>-(j)ian*, *·rán>-arn*, *·rân>-(r)ian*, *·rjân>-ern*, *·rjân>-(r)ian* (where raised *j*, *r* and *rj* stand for features of the initial and/or final consonants that gave rise to *j*-, *r*- or combined *rj*-umlauts). In Type A syllables the umlauts were postnuclear, turning nuclear *a* into front, retroflex or (in the case of *w*-umlauts, not illustrated above) back-rounded vowels. Final consonants were also sometimes affected. For example, under the influence of *j*-umlaut, final *-q* in Category XI and final *-k^w* in Category IX were palatalized e.g. 笛 *dek<*dec<*jśk^w*, 激 *kek<*kec<·jśq*. In Type B syllables the umlauts affected primarily the medial rather than the nuclear vowel, though in some cases both medial and nuclear vowel were affected. At the end of this article an illustrative table is given to show how the *Ch'ieh-yün* rhymes developed out of the *Shih-ching* rhyme categories.

The prosodic contrast that led to the separation into Type A and B syllables had marked effects on the initial consonants as well as the rhymes. In Middle Chinese velar initials had back allophones in Type A syllables and front allophones in Type B. The voiced stop, *g*, had only a front allophone and was confined to Type B. It was in complementary distribution to the fricative *γ* which had no front allophone and appeared only in Type A. Old Chinese dental stops became palatal affricates in Type B syllables and the nasal *n* became palatal *ɲ*. Correspondingly Old Chinese **l* was palatalized to *ʎ* or *j*

in Type B and was hardened to a stop, *d*, in Type A. Old Chinese **l'* similarly became *ɬ* in Type B and *t'* in Type A. A full discussion must be left for another occasion.

(b) Tones

The origin of the rising and departing tones of Middle Chinese from final glottal stop and *-s* respectively has been discussed elsewhere and will be assumed here without further argument (Pulleyblank 1962, 1973b, Mei 1970). The ultimate origins of *-s* and *-ʔ* may actually be more complex but this reconstruction is sufficient for the Old Chinese period. *-s* could be added to other finals of all kinds, including stops, which accounts for the *hsieh-sheng* and rhyming contacts between departing and entering tones. **-ps* (equivalent to most cases of *-b* in Karlgren's reconstruction) early assimilated to **ts-*, as shown by the rhyming of words like 𪛗 *noʃh* < *náps* (cf. 納 *nap*) in Category IIIa in the *Shih-ching*. By Han, or perhaps even much earlier, **-ts* had become *-s* (merging with original **-s* ?). This *-s* survived as late as the 6th century A.D. (Pulleyblank 1973b). **-ks* seems to have become a velar fricative by Han times, though evidence is less abundant than for *-s*. After nasals *-s* had probably become *-h* by Han times.

Even at the time of the *Ch'ieh-yün* it seems probable that the rising and departing tones were characterized as much by final laryngeal features as by pitch and contour and I therefore use raised *ʔ* and *h* as markers of these tones for Early Middle Chinese.

(c) Yin finals

Apart from the reconstruction of *-s* after stops, which accounts for many of the *yin* finals in departing tone, I reconstruct voiced continuants in the various *yin* rhymes (Pulleyblank 1962, 209 ff.). As the final dental in IIIa and IV(ii)a I reconstruct **l*. This replaces **δ* which I postulated in 1962 both initially and finally to correspond to Sino-Tibetan **l* (Middle Chinese *l* being cognate to Sino-Tibetan **r*). I have since concluded that, as this phoneme, when initial, was used for foreign *l* even in the Han dynasty, it should be so reconstructed for Old Chinese.

In final position **l* became *-j* after *ə* (IIIa) very early. This is

shown by the fact that IIIa and Va were already merging in *Shih-ching* rhyming. More will be said on this point below. **-aI* must have similarly become *-aj* at a somewhat later date. It did not immediately merge with original **aj* (VIa) however, presumably because the latter had meanwhile fronted to *ɛj* and perhaps monophthongized to *ɛ* (see below). There is occasional rhyming contact, however, in Late Chou times. The value *-aj* for IV(ii)a is implied by reflexes of this category in colloquial Min.

	EMC	Fu-chou	Amoy	Ch'ao-chou	Lung-tu
我	ŋaʔ	ŋuai	gua	ua	wǎa (cf. Hakka ŋai)
舵	da	tuai	tua	tua	(cf. Canton. t'ai)
破	p'a ^h	p'uai	p'ua	p'ua	phua
麻	maɿ	muai	muã	mua	mua
沙	saɿ	sai	sua	sua	sua
火	xwaʔ	xuei	he	hue	fuaj
坐	dzwaʔ	soey	tse		soj
倚	?iãʔ	ai	ua	ua	
蛇	ʒiaɿ			tsua	sua

(EMC stands for Early Middle Chinese. Note that in Southern Min the glide *-j* has been lost but this is true also for words which have *-j* in Middle Chinese, like 賴 *laɿ^h* Amoy *lua*.)

One also finds forms in *-j* in early Vietnamese and Korean loans. Haudricourt (1954, 363) gave the following examples from Vietnamese: *ngài* "moth" 蛾 *ŋa*, (Sino-Viet. *nga*), *mài* "grind" 磨 *ma* (SV *ma*), *ngói* "tile" 瓦 *ŋwaɿ²* (SV *ngōa*) *do'i* "change" 移 *jiã* (SV *di*), *cõ'i*, *cũ'i* "ride" 騎 *giã*, *giã^h* (SV *ky.*); *gõ'i*, *gũ'i* "entrust" 寄 *kiã^h* (SV *ký*).

In Korean we have *t'aj* [t'æ] for 馱 *da*, *maj* [mæ] for 磨 *ma*, *oaj* [wæ] for 倭 *?wa*, *t'aj* [t'æ] for 妥 *t'wa²* (Kōno 1968, 87–90).

The loss of the final *j* glide which gave rise, in Type A syllables, to the *Ch'ieh-yün -a* (歌) rhyme, took place in Northern Chinese during the Later Han period, for by the end of the second century such words were being used to write open *-a* syllables in Buddhist transcriptions: 阿 *?a=a*, 羅 *la=la*, *ra*, etc. This change did not affect proto-Min, however.

The *yin* final of the velar categories VII and VIII will be written here as γ (replacing $-h$ of my previous article). At least by the Han period this must have been pronounced with very little friction and might be better designated as the central unrounded semivowel $\dot{\gamma}$ which I have postulated as a constituent of vowel systems (Pulleyblank 1969 and 1972). This would be realized phonetically as a retraction and lengthening of a preceding nucleus, $[\dot{\gamma}] = [a]$. In VIII such finals in level tone were normally used during Han to transcribe foreign open *a*-syllables: 烏 $\gamma\gamma < * \gamma \dot{\gamma} i$ $[\gamma a] = a$, 奴 $n\gamma < * n \dot{\gamma} i$ $[n a] = na$, 蒲 $b\gamma < b \dot{\gamma} i$ $[b a] = ba$, etc. Sino-Tibetan cognates also show open *-a*, as in 五 $\eta\gamma^2$ “five,” Tib. $\eta\eta a$, 苦 $k'\gamma^2$ “bitter,” Tib. $k'a$, 魚 $\eta\dot{\gamma}\ddot{a} < * \eta \dot{\gamma} \dot{\gamma} i$ “fish,” Tib. $\tilde{n}a$, Burm. ηa . Compare also early loanwords in Southeast Asian languages like Thai *ha* “five”; Khmer *sna* “crossbow” = 弩 $n\gamma^2$; Vietnamese *mả* “tomb” = 墓 $m\gamma^2 h < * max$.

After the higher nuclear vowel, ∂ , in VII there is more evidence of an overt consonantal final. To explain the forms in early Tai loans, F. K. Li (1945, 341) postulated an intermediate stage in which $*g$ (accepting Karlgren's Archaic reconstruction) first became γ , then u , equivalent to my $\dot{\gamma}$, or better, the corresponding glide $\dot{\gamma}$. Whether this is really an intermediate stage in the loss of an original stop or is to be projected back unchanged to Old Chinese is another question for which there is little evidence at present.

The *yin* finals corresponding to palatals and labiovelars will be reconstructed as $-j$ and $-w$ respectively and the uvular final in XI will be $-x$. These reconstructions will be discussed more fully below.

Final Palatals

The basis for Karlgren's reconstruction of the same front vowel, *e*, in the two rhyme categories V and VI with dental and velar finals respectively is that the Middle Chinese reflexes of these categories have front vowels and these types of final consonant in the *yang* (nasal) and *ju* (stop) rhymes. There are at least two serious objections one can make against this hypothesis. In the first place there is no apparent reason why there should not also be a category with this vowel in front of labial finals. In the second place the way in which the

hypothetical original front vocalism has developed in the two categories from Old to Middle Chinese is quite different. V merges with Category IV (*əl*, *ən*, *ət*) and develops as an “inner” rhyme category while VI shows most contact in its development with VIII (*əɣ*, *əŋ*, *ək*) and develops as an “outer” rhyme category. If one adopts Karlgren’s reconstruction (or F. K. Li’s, which replaces Karlgren’s *e* by *i* before the same final consonants) one is forced to assume that there was a drastic lowering of the vowel before velar finals or a raising before dental finals, or both. There is not the slightest evidence that such shifts took place. Nor does it seem probable that final dentals or velars would have had such effects in view of the fact that the central vowels, *ə* and *a*, remained quite distinct in front of these same finals.

Both difficulties are disposed of at the same time if we hypothesize that Categories V and VI, instead of having the same vowel and different finals, had different vowels, *ə* and *a*, and the same finals, namely palatals. After the higher nucleus in V, the palatal stop and nasal were advanced and merged with the dentals: *əŋ* > *en*, *əc* > *et*. After the lower nucleus in VI they were retracted and merged with the velars: *əŋ* > *ɛŋ*, *əc* > *ɛk*. In both cases the original palatal feature was transferred to the vowel.

The merger of the *yang* and *ju* rhymes of V with those of III took place between the time of the *Shih-ching*, when they were quite distinct, and the Han dynasty, when they had completely merged in poetic rhyming. The merger of the corresponding *yin* rhymes was even earlier and must have taken place in the opposite direction, that is, by the shift of *-əl* to *-əj*. As already noted, they are only imperfectly distinguished in *Shih-ching* rhyming and were not separated by the Ch’ing philologists or, in consequence, by Karlgren. Some twentieth century linguists like Tung T’ung-ho and Wang Li have tried to establish a distinction on the basis of *hsieh-sheng* relationships as well as rhyming but uncertainties remain as to the correct assignment of particular words.

Category VI did not merge with VIII but in Eastern Han we find that certain words from VIIIb had moved over to VI, namely those that were later placed in the 庚 *keng* rhyme in the *Ch’ieh-yün* like 京 *kieŋ*, 兵 *pieŋ*, 行 *ɣeŋ*, 猛 *meŋ*? Available evidence is hardly sufficient

to tell with certainty what this means in terms of phonetic change but it is obviously easier to account for if we assume that it involved a fronting of $-\eta > \eta$, than if we think merely in terms of the kind of small alteration in the vowel that is allowed for in Karlgren's, or F. K. Li's, system. According to Li, $-a\eta$ changed to $-o\eta$ under the influence of r as the second element in an initial cluster; at the same time $-jia\eta$ and $-lja\eta$ changed to $-jo\eta$. It is not obvious how either of these supposed shifts makes sense in terms of the features involved or why the low central vowel in $-o\eta$, $-jo\eta$ should have rhymed with the front vowels in $-\varepsilon\eta$, $-je\eta$, $-ie\eta$ (Category VI) rather than the central vowel in $-a\eta$, $-ja\eta$ (Category VIII). Even if we decline to take seriously the exact values of the vowels in $-o\eta$, $-jo\eta$, it is still unexplained why such a shift in rhyming should have affected the $-a\eta$ group and not the $-an$ and $-am$ groups.

What appears to have happened in fact is that an r -feature in the initial complex was thrown forward on to the final and caused an assimilatory fronting of the final velar nasal: $-ʔa\eta > a\eta\eta > a\eta\eta$. In Type A syllables this happened after initials of all types and, as far as the r -umlaut is concerned, is parallel to what happened in all other rhyme categories: $-ʔam > a\eta m$, $-ʔan > a\eta n$, etc. But in the comparatively lax rhyming practice of the Han dynasty it was evidently possible to rhyme plain and retroflex vowels: $-an$, $-a\eta n$, etc. It was only with the more exacting rhyming standards that were established in the late fifth century that they were kept apart. In the case of final $-\eta$, however, not only did the vowel become retroflex but the final consonant also received a fronter articulation and this was reflected in rhyming practice.

A complete assimilation of $-\eta$ to a preceding r -feature should have given a final retroflex nasal, η , but η and η are close to one another in place of articulation and we must suppose that the difference in manner, if it existed, was ignored as phonologically non-distinctive. This is similar in principle to the phonemic merger that occurred between retroflex and palatal initials between Early and Late Middle Chinese.

A unique feature of the r -umlaut in VIII as compared to other rhyme categories is that the final was affected in some Type B syllables

as well as Type A, namely those with velar (including labiovelar) and labial finals, e.g.: **krànŋ* > *kriən* > *kiarŋ* > *kiarŋ* (later becoming Early Middle Chinese *kieŋŋ* and Early T'ang *kieŋ*) (cf. 涼 *liən*); but 張 **trànŋ* > *trianŋ*. The condition under which the retroflex feature was thrown forward on to the vowel seems to have been the change of *-ri-* to *-ri-*, followed by the loss of *r*: *-ri-* > *-i-*. After dental and palatal (later dental sibilant) initials this did not take place because the initials themselves absorbed the following *r* and became retroflex: *tr* > *tr* (still written *tr* in my Middle Chinese reconstruction but structurally a single segment, not a cluster), *cr* > *tʂ*. This phenomenon deserves a fuller treatment but the general principle seems sound.

During Later Han *r*-umlaut seems only to have affected the nasal *-ŋ*, not the stop *-k*, as far as one can tell from rhyming. Long before the *Ch'ieh-yün*, however, the same changes had taken place in the entering tone, e.g. 格 *krák* > *kaŋc* > *keŋk*, 白 *brák* > *baŋc* > *beŋk*, 戟 *kràk* > *kiarŋc* > *kiek*.

Assimilation of *-k* to *-c* involving a transfer from VIII to VI also occurred in the post-Han period in Type B words with dental sibilant (Old Chinese palatal) and palatalized dental (Middle Chinese palatal) initials, e.g. 石 **d'ák* > **dʒiak* > *dʒiac* > *dʒiek*, 迹 **càk* > *ciàk* > *tsiàc* > *tsiek*. In this case the first stage of the change must have been fronting of the medial *ɨ* to *i*, after which *-k* underwent palatal assimilation. Final *-ŋ* was not affected in this case.

The palatal character of the finals in VI was still strong in Middle Chinese as M. Hashimoto has emphasized in a number of articles. He, in fact, wants to reconstruct final palatals there in Middle Chinese. At least by the time of Late Middle Chinese, it seems clear to me that the palatal finals were phonologically identified with the velars and the palatal feature had become transferred to the vowel (Pulleyblank 1970–71), but there is no doubt that Hashimoto's evidence and arguments support the idea of palatals as the original finals in this category.

The development of the *yin* final *-aj* from Old Chinese to Middle Chinese may also be briefly sketched. By Han times *-aj* must have been fronted to [ɛj] and then monophthongized to [ɛ]. (Compare the change of Middle English *-aj* to [ɛ] in Early Modern English in words

like *day*, or the development of *-ai-* to [ɛ] in French.) It therefore remained distinct from the new *-aj* that developed out of *-ai* in IV(ii)a. By Later Han, however, Type B syllables from VIa had become /*(j)iã*/, phonetically [iɛ] and Type B syllables from IV(ii)a had also lost their final glide and become /*iä*/. This led to a merger which created the *Ch'ieh-yün* 支 rhyme. Later Type A syllables /*ái*/ [ɛ] were raised to [e] and diphthongized to [ej], merging with Type A finals from V (including some words that had originally been in III) and giving the *Ch'ieh-yün* 齊 rhyme.

The argument for reconstructing palatals in V and VI has so far been based on internal reconstruction. That is, it has been shown that such a reconstruction gives a much simpler and linguistically more convincing account of the distribution of the rhymes in Old Chinese and of their development to Middle Chinese. Further support can be given from *hsieh-sheng*, rhyming and word family contacts between the two categories but since such evidence, while consistent with the hypothesis of final palatals, might be consistent with other interpretations as well, I shall not stress it here.

In addition to this internal evidence, there is some important comparative evidence from Burmese, which still had the palatal finals *-ñ* and *-c* when it was first reduced to writing. It appears that those working on comparative studies in the Tibeto-Burman field mostly assume that the final palatals of Written Burmese are a secondary development arising from the palatalization of dentals and velars after *-i*. As far as I can tell, however, the opposite hypothesis has not been seriously tested. In some cases it is no doubt true that such palatals are of secondary origin. For instance, the word *hrac* "eight," written *het* on the Myazedi inscription, must have originally had final *-t*. Cf. Tibetan *brgyad*, Chinese *pert* < **prjāt*. This does not exclude the possibility that there may have been a distinct series of palatal finals also. The absence of such consonants in modern languages of the area is not a very significant argument, since there the long term trend throughout the whole of East Asia has been for the reduction of distinctions at the end of the syllable. In fact this persistent tendency in itself provides an argument against the idea that Burmese may have created a new series of palatal finals at one stage and then lost them again. This does not mean, of course, that at a time when

a distinct series of final palatals existed other types of final consonant might not have been palatalized under various influences and merged with the existing palatals, as happened in Chinese in the various shifts from VIII to VI that have been described above.

Certainly it is very striking that, apart from the word for “eight” noted above, wherever we can find good cognates in Chinese for Burmese words spelt *-ac*, *-añ*, they fall in either Category V or VI.

Burmese	Chinese	Tibetan
sac “new”	新 <i>sin</i>	
sac “wood”	薪 <i>sin</i>	śiŋ
hnac “two”	二 <i>ɲij^h</i> < *nə̌ɰ	gnyis
ʔachac “joint”	節 <i>tset</i>	ts’igs
ʔac “squeeze, throttle”	噎 <i>ʔet</i> “choke”	
ʔac “a unit, one”	一 <i>ʔjit</i>	
chac “cut in parts”	切 <i>ts’et</i>	
hnac “year”	年 <i>nen</i>	niŋ
mañ “name”	名 <i>mjieŋ</i>	miŋ
tañ “straight (up)”	正 <i>teiɛŋ^h</i>	
mrañ “to sound”	鳴 <i>miɛŋ</i>	

There are also some good correspondences between Va *-əj* and Tibeto-Burman words with final *-i* or *-j*.

Chinese	Burmese	Tibetan
死 <i>sij[?]</i> “die”	<i>se</i> < *səj	ñč’i-ba, perf. śi
屎 <i>ɛij[?]</i> “excrement”	ʔakhye:	lči-ba < *lhyi

Contrast Tibetan *-l* for Chinese *-əl* (IIIa) in 洗洒 *sej[?]*, *sen[?]* “wash,” Tibetan *bsil-ba* “wash.” (Schuessler 1974 gives several other examples but in many of them the phonetic and semantic correspondences are not close enough to be convincing without further evidence. Note that Schuessler following Karlgren, fails to distinguish IIIa and Va and is led to the unlikely hypothesis that Sino-Tibetan *-y* [read *-j* in my notation] corresponds to Chinese *-l[?]* or *-lV* [where V is any vowel].)

Chinese **-aj* (VIa) can be found in 舐 *ɬiə̌* “lick” < **laj[?]* (compare the colloquial Cantonese readings *lai*, *sai*, both readings in upper rising

tone): Tib. *lč̣e* < **lhaj* “tongue” (Benedict reconstructs Sino-Tibetan *s-lay*, *m-lay*).

As a final argument in favour of reconstructing final palatals for Old Chinese, I wish to identify them with a corresponding set of initial palatals. As we shall see below, an important argument for reconstructing final labiovelars is that we also require such consonants initially. This is also true of the palatals. The palatals of modern Mandarin have, of course, developed in quite recent times from velars and dental sibilants in front of high front vowels or glides. Middle Chinese palatals, which have become retroflex or plain dental sibilants in Mandarin, had similarly developed since the Old Chinese period, mainly from dental stops, partly from velars. There is still, however, the Middle Chinese dental sibilant series to account for. In some cases, as Bodman and I have suggested, these initials may have arisen from the metathesis of *st*, etc. On the other hand there are good equations to show that in the main this series corresponds to the Tibetan dental affricates, *ts*, *ts'*, *dz*, and to Burmese *c* and *ch*, now pronounced [s] and [s']. Though in no case is the modern pronunciation palatal, the way in which these initials were written when Burmese was reduced to writing indicates a palatal pronunciation at that time. The corresponding initials in Mon, from which the Burmese borrowed their alphabet, are still palatal stops and even in Burmese the corresponding nasal *ṇ* is still pronounced as a palatal. If my equation of Burmese final *-c* and *-ṇ* with Old Chinese final palatals is correct, this is a formal link with Burmese initial *c*, *ch* and hence with Tibetan and Chinese *ts*-, *ts'*-, *dz*-.

The original Sino-Tibetan palatals must have shifted forward to dental articulation independently in Chinese, Tibetan and Burmese under the pressure of new palatals of different origin. This is a process for which one can find many parallels in other languages, for example in Slavonic and Old French.

As I showed in my article on the “Consonantal System of Old Chinese” (1962, 109) the Middle Chinese dental affricates were used to transcribe foreign palatals during the Han dynasty, before the Middle Chinese palatals had appeared. Thus we have 龜茲 *kuw-tsi*=*Kuča*, 且末 *ts'äṛ-mat* Calmadana, 子合 *tsi²-γap* for the place later transcribed

as 朱駒波 *tsuă-kuă-pa* (Sung Yün) and 斫句迦 *tsiak-kow-kiă* (Hsüan-tsang), 丘就却 *k'uw-dzuw-kiak* (for 劫 *kiăp*?, see Pelliot 1914, 401), 師子 *si-tsi*² "lion", cf. Tokharian B *ṣecake*. This consistent use of Middle Chinese dental affricates for foreign palatals in Han does not, of course, prove that they were palatals at the time since, if the Middle Chinese palatals had not yet developed, dental sibilants might have been the only Chinese phonemes available, but obviously the sound equation is much better if we have grounds to believe that the Middle Chinese dental sibilants were actually palatals in Han times.

Final Labiovelars

The fact that in many Middle Chinese rhymes one finds a contrast between *k'ai-k'ou* and *ho-k'ou* only after velar or laryngeal initials is best accounted for by reconstructing labiovelars as a distinct class of initials in Old Chinese (Pulleyblank 1962, 95). This further suggests that one can account for one or more of the three *Shih-ching* rhyme categories that gave rise to back vowels followed by velars in Middle Chinese by reconstructing final labiovelars. Proposals of this kind have, in fact, been made in the past by a number of scholars.

Rai (1953, 1957) proposed labiovelars or uvulars in all three. Wang Li (1958) proposed diphthongs, *əu* and *au*, for the *yin* rhymes of IX and XI, and these same diphthongs followed by *ŋ* and *k* for the corresponding *yang* and *ju* rhymes. Since such diphthongs cannot occur in front of labial or dental finals, this is more or less equivalent to positing final labiovelars. This was essentially the solution I adopted in 1963. F. K. Li (1972) has now published his new system which has labiovelar finals in these same two categories. Chou Fa-kao (1969) has labial glides plus velars in categories IX, X and XI.

The solution which I now wish to put forward differs from my previous one and from that of Wang Li and F. K. Li in reconstructing labiovelars in IX and X instead of IX and XI. Category XI, which differs from the other two in having no final nasals, requires a different solution which will be discussed in the next section.

One source of confusion has been Karlgren's assumption that Category X had the highest of his three back-rounded vowels, *u*. In this way he projects back into Old Chinese his Middle Chinese reconstruction for the Type A (Grade I) rhymes derived from this category. The corresponding Type B rhymes, on the other hand, have a lower vowel in Early Middle Chinese. I reconstruct them as *-uɔŋ*, *-uɔk*, *-uǎ[uɔ]*. Karlgren has *-iʷong*, *-iʷok*, *-iʷu*. This is appreciably different from mine in terms of vowel height only in the *yin* rhyme. This is not the place for a full discussion of Early Middle Chinese values but one may note the existence of early Vietnamese loans such as the following which support my reconstruction: *búa* "axe" = 斧 *puǎ*² (SV *phǎ*), *chúa* "master" = 主 *tsuǎ*² (SV *chu*), etc. (see Haudricourt 1954, 360 for these and other examples). Similar forms are also found in Min colloquial, e.g. 斧 Fu-chou *p'uɔ*, 朱 *tsuǎ* Fu-chou *tsuɔ*, 鑪 *tsuǎ*^h Fu-chou *tsuɔ*, 厨 *druǎ* Fu-chou *tuɔ*, 句 *kuǎ*^h Fu-chou *kuɔ*. This Middle Chinese rhyme actually contains not only Type B syllables from Xa but also *ho-k'ou* words from VIIIa. This merger, like the various mergers of words from Category VIII with Category VI, is *prima facie* evidence of a low nuclear vowel.

The development of Category X, which I assume to have had labiovelar finals after the low nuclear vowel, *a*, was in fact quite parallel to that of Category VI, as can be seen by the following comparison. (For the sake of simplicity I omit the Grade II rhymes, which developed under the influence of *r*-umlaut.)

	Type A	Type B
VI	áŋ > ɛŋ > eŋ	àŋ > iɛŋ
	ác > ɛk > ek	àc > iɛk
	áj > ɛ > ej	áj > (j)iǎ
X	áŋ ^w > ɔŋ > oŋ	àŋ ^w > uɔŋ
	ák ^w > ɔk > ok	àk ^w > uɔk
	áw > ɔ > ow	àw > ɔ > uǎ

During the Han dynasty we must assume that the *yin* final had reached the stage of monophthongizing to *ɔ*. At this period it rhymed freely with VIIIa, which had a low back unrounded vowel [*a*] (see above). There was no actual phonological coalescence except for the

shift of *ho-k'ou* words of Type B from VIII to X that we have already noted. The rhyming of [ɔ] and [a], although not perfect, was evidently close enough by the comparatively lax standards of the time. Characters from Xa appear comparatively rarely in transcriptions of the period while characters from VIIIa are very common. This discrimination in itself argues against the idea that there was an actual phonological coalescence. A phonetic value [ɔ] fits well in general for the transcription value of Category X as a whole at that period. I will here cite only the example of 高附 *kaw-buǎʰ* < **kaβ-box* = *κάβουρα*, i.e. modern Kabul (Pulleyblank 1962, 223).

Just as Category X was parallel in its development to Category VI, so Category IX was parallel to Category V, as far as Type B syllables are concerned.

V	əŋ > in	IX	əŋ ^w > uŋ
	əc > it		ək ^w > uk
	əj > ij		əw > uw

The parallel breaks down, however, in Type A syllables. In V the vowel became tense and remained high: *ej*, *en*, *et*. In IX it remained lax. The *yin* final -əw dropped to -əw. We must suppose that this also happened in the *yang* and *ju* finals: -əŋ^w > -əwŋ > -əwŋ, -ək^w > -əwk > -əwk, after which there was monophthongization to -ɔŋ, -ɔk.

These developments resulted in a curious crossing over between Categories IX and X, so that in Middle Chinese Type A rhymes of IX had come to rhyme with Type B rhymes of X and vice versa.

Type A	Type B
oŋ < əŋ ^w (X)	uŋ < əŋ ^w (IX)
ɔŋ < əŋ ^w (IX)	uɔŋ < əŋ ^w (X)

Note that, though phonologically distinct, *oŋ* and *uŋ* were sufficiently similar to rhyme together in the *Ch'ieh-yün* and were treated as parts of the same rhyme. The corresponding *yin* rhymes, 侯 -ow and 尤 -uw, were placed in separate rhymes in the dictionary but rhymed together freely in poetry. The crossing over we have noted may be compared to what has happened in Cantonese to the Late Middle Chinese rhymes

-*aw* and -*əw*, which have become *ow* and *aw* respectively, reversing their Middle Chinese positions in terms of height.

Something must be said about the close relationship between the *yang* rhymes of Categories I(*əm*) and IX(*əŋ^w*). There are frequent rhyming contacts between the two categories in the *Shih-ching* and Wang Li took this as evidence that IXb actually ended in -*m*. On the whole the two categories are separate, however, and such an assumption is not necessary to account for occasional rhyming contacts. The final labiovelar consonant in IXb may have been pronounced like the final in Vietnamese -*ôŋg*, which is realized with double, labial and velar, articulation [*əuŋm̐*] (Henderson 1966, 166). Such double articulation is also found in Min dialects which have -*w* diphthongs followed by velar finals. Such rhyming contacts in the first half of the second millennium B.C. are to be distinguished from some actual shifts from I to IX that occurred during the Han dynasty as a result of dissimilation of the final labial from a labial initial. The best known case is that of 風 *puŋ* "wind," which has 凡 *buəm* as its phonetic and which still rhymed in -*m* in Former Han times. As late as 200 A.D. the *Shih-ming* records two pronunciations, one in -*m* and one in -*ŋ*, in different geographical areas (Bodman 1954).

Final Uvulars

Category XI differs from IX and X in a number of obvious ways. In the first place it has no nasal finals. If, as some have done, we match it with IX as having labiovelar finals, we are left with this gap in distribution to account for. Wang Li's solution, which derives the nasal finals of IX from -*wəm*, removes the necessity of reconstructing final labiovelar nasals but since there is a need to reconstruct such a nasal initially along with the labiovelar stops, it does not really solve the problem. Other reasons for rejecting this solution have been given above.

As well as showing no nasal finals, XI shows much less evidence in its later development of *w*-features than do IX and X. The *yin* rhymes end in -*w* in Middle Chinese but, as we shall see, this can be

explained as a secondary development from a bilabial fricative $-\beta$. Unlike the Grade III rhymes which develop out of IX and X, those which develop out of XI are $k'ai-k'ou$ ($-i\check{a}w$ and $-iak$) and they do not lead to the development of labial fricatives from labial stop initials. Contrast 表 LMC and EMC $pi\check{a}w^2$ (XI) with 缶 LMC fuw^2 EMC puw^2 (IX) and 府 LMC $fu\check{a}^2$ EMC $pu\check{a}^2$ (X). Type A entering tone syllables often show no lip-rounding in Middle Chinese, e.g. 樂 lak , 鑿 $dzak$, 鶴 γak , 爆 pak . On the other hand the Grade II rhyme (r -umlaut) is $-ɔrk$, merging with the Grade II rhymes derived from IX and X and some words in Grade I also show rounded vowels, e.g.: 霍 $\gamma\check{ok}$, 暴 bok , 瀲 $l\check{ok}$ (as well as lak and lek). These points suggest finals for XI with a certain tendency to cause rounding of the preceding vowel but without lip-rounding as an inherent feature of the consonants themselves.

The fact that we could readily identify classes of initial consonants corresponding to final palatals and labiovelars was of great assistance in establishing the probability of such a reconstruction. This is not so easy in the case of uvulars but there are some indications that would favour the hypothesis that some Middle Chinese velars were derived from a distinct class of uvulars. Haudricourt (1954) made such a suggestion on the basis of early Thai loans which show uvular initials. He mentioned in particular Thai qan "steel" = 鋼 kay , and Thai $G\text{am}$ "gold" = 金 kim . What is needed, of course, in order to make a successful reconstruction of uvulars in Old Chinese is to find some criterion by which one could distinguish the uvular and velar *hsieh-sheng* series. One may tentatively suggest that uvulars might be found where we find velars alternating with glottal stop, as in 可 $k'a^2$ 阿 $2a$, 甲 $kapp$ 押 $2app$, 今 kim 陰 $2im$. Original velars might be indicated, on the other hand, where we find velars alternating with d or j , as in: 庚 $kerr$ 唐 day , 谷 kok , $juck$, 姜 $kiaŋ$ 羊 $jiay$, 姬 kj 匣 $j\check{i}$, etc. (I was formerly inclined to reconstruct $-l$ - clusters in such words but this was based merely on an analogy with the development of Old Chinese $*l$ and was not supported by comparative evidence.) These suggestions can only be made tentatively at present pending further study.

Another indication that Old Chinese may have distinguished uvular and velar initials may be found in the allophonic distinction which existed in Early Middle Chinese in the pronunciation of velar initials

before a mid or low vowel (Type A syllables) on the one hand and before a high vowel (Type B syllables) on the other. In the former situation velar initials had a back velar or uvular pronunciation, while in the latter situation they were true velars. This is shown, among other things, by the exclusive use of Type B syllables to represent Indian velars in most pre-T'ang Buddhist transcriptions (Pulleyblank 1965). It also provides an explanation for the fact that in Early Middle Chinese *g* occurs only in Type B syllables and is replaced by *γ* in Type B syllables (*ɦ* in the southern variety of Early Middle Chinese). A voiced uvular stop is difficult to produce and normally tends either to become unvoiced or to become a fricative. This contrast is allophonic in Early Middle Chinese but it could well represent a stage in the merger of originally distinct uvular and velar consonants.

The voiced uvular final of Category XI may be written with the IPA symbol representing a voiced uvular fricative or frictionless continuant, analogous to velar *γ* (or *ɣ*), labiovelar *w*, palatal *-j*, and dental *-l*. By Han times, however, the evidence points to a bilabial *β*. Such a shift from an uvular to a labial fricative may be compared to the replacement of Middle English *gh* [χ] by Modern English [f] in words like *laugh*, *cough*, etc. Evidence for a bilabial articulation is found in Han dynasty transcriptions (Pulleyblank 1963, 206). The shift from a bilabial to a labiovelar pronunciation must have occurred in early post-Han times.

The absence of an uvular nasal final is not surprising. Such a phoneme is possible but must be comparatively rare. Common Thai had no uvular nasal. The distributional gap would be even better accounted for if one supposed that the uvulars had earlier had a laryngeal or pharyngeal pronunciation but I have no evidence at present to support such a conjecture.

Another distributional gap that must be discussed briefly is the absence of a twelfth rhyme category with uvulars following the higher nuclear vowel *a*. Word family connections between XI and IX such as 學 *ɣɔrk* “study” (IX), 校 *ɣarw^h* < **-áqs* “school” (X), 療 *tr'uw*, *lew* “be cured, recover” (IX), 療 *liǎw^h* “cure, heal” (XI), 繆 *luw* “tie round, strangle” (IX) (cf. also 繆 *mjiw* “bind around”), 繆 *liǎw²*, *lew²* “bind around, wrap” (XI), 簫 *sew* “pan pipe” (IX), 簫 *sew* “pipe, flute” (XI),

蠅 *sew* “spider” (IX) 蛸 *sew* “spider” (XI), suggest that *-əq*, *-əʃ* early shifted to *-əkʷ*, *-əw*. (Note that there are also word family connections between IX and X.)

* * *

I shall touch only briefly on the question of comparative evidence for final labiovelars and uvulars. Much more study will be needed to establish the correct correspondences. As far as labiovelars are concerned it is relevant to note that, matching its final palatals, Written Burmese has velar, but not dental or labial, finals after the vowels usually transcribed as *ui* and *o*. These vowels can be analyzed as historically derived from */əw/* and */aw/* (Pulleyblank 1963). There is thus a structural symmetry between final palatals and final labialized velars or labiovelars like that found in Vietnamese and the Fu-chou dialect of Chinese. There are also correspondences between these finals in Burmese and the final labiovelars reconstructed for Chinese. Two undeniably good cognates are the words for “six” and “nine”: 六 *luk* < **-rəkʷ*, Burm. *khrok* /k'rawk/, Tib. *drug*; 九 *kuw*² **kəw*², Burm. *kui* /kəw/, Tib. *dgu*.

I know of no case of contrasting velar and uvular finals in present day Sino-Tibetan languages. Compare, however, the word for “eye” with the word for “six” quoted above. In Chinese 目 *muk* also comes from Category IX and rhymes with “six” also in Middle Chinese. Tibetan *mig* and Burmese *myak*, however, show a quite different correspondence, with no trace of a back vowel. (For other Tibeto-Burman forms see Benedict 1972.) We could account for this contrast by reconstructing **mjəq*. The word is probably cognate to 覿 *mər̥k*, *mər̥wʰ* * < **-əq(s)* “delineate; face, appearance.”

Ho-k'ou after Dental and Palatal Initials

The limited occurrence of the *k'ai/ho* distinction (absence or presence of a back-rounded medial) except after velar initials has been cited above as a reason for positing a distinct class of labiovelar

initials in Old Chinese. One must still account for the existence of such a distinction after dental and palatal initials in Middle Chinese rhymes with dental finals or derived from dental finals. Yakhontov (1960) used this as the basis for reconstructing back-rounded vowels, *u* and *o*, before dental finals in Old Chinese and this was also my own first opinion (Pulleyblank 1962, 142). Having now rejected this hypothesis, it is still necessary to account for the phenomenon in another way.

Part of the answer can be attributed to the palatalization of labiovelar x^w before *-i-* in Type B syllables. This must be the explanation, for example, for 恤 *swit* < $*x^w\acute{a}c$, with 血 *xwet* < $*x^w\acute{a}c$ as phonetic. If x^wi- became palatalized to ɕ^wi- during the Old Chinese period, we should expect it to become *swi-* in Middle Chinese in the same way that the original palatal stops became dental affricates: $*c > ts$. Other examples of *sw* < $*x^w$ are 荀 *swin* (cf. 洵 *xwen^h*); 戊 *swit*, phonetic in 歲 *swiä^h* (cf. 薺 *xwaj^h*, etc.); 宣 *swiän* (cf. 喧 *xwiän*); 攄 *swiät* (cf. 噤 *xwej^h*); 嵩 *swiä²* (cf. 攜 *ɣwej^h*). One also finds the corresponding voiced sound, *zwi-*, in this type of *hsieh-sheng* series, presumably representing an earlier ɕ^wi- < γ^wi- , e.g.: 旬 *zwin*, 攄 *zwij^h*, 穗 *zwij^h* (cf. 惠 *ɣwej^h*), 還 *zwiän* (also read *ɣwajⁿ*). The assumption that *zwi-* represents an earlier ɕ^wi < γ^wi will help to account for a curious phenomenon noted by Paul Pelliot in some early Buddhist transcriptions where Middle Chinese *zwi-* stands for Sanskrit or Prakrit *vi* or *ve* (Pelliot 1933). Pelliot was puzzled by it and inclined to attribute it to a peculiarity of some unknown Central Asian language or dialect but it probably reflects a Chinese dialect in which γ^w had failed to palatalize.

Other cases of *ho-k'ou* with dental or palatal initials may have arisen from dental suffixes added to words ending in *-w*. Thus 脫 *t'wat* < $*l'wät$ "take away, strip" is no doubt related to 偷 *t'ow* < $*l'aw$ "steal" and could be derived from $*l'aw + t$, with metathesis of *w* in front of the nuclear vowel required by rules of syllabic structure. Similarly one can derive 栗 *ɳwiän²* < $*nwän²$ "soft, weak" from $*naw + n$. Compare 懦 *nuä* < $*naw$ "weak, timid."

In still other cases it seems very likely that the labial medial may be derived from a lost labial or labiovelar initial consonant. E.g.

赫 lwan, 擊 lwiän: 變 piän^h < *pràns, 變 marn < *mrán
 律 lwit, 聿 jwit: 筆 pit < *prət
 論 lon, 倫 lwin: 論 kon^h < *kw-

That is, when an initial labial or labiovelar was lost in a cluster of the type *Pr-*, *Kwr-* so that medial *r* became initial, changing to *l* in Middle Chinese, the labialization of the original initial was retained as a medial *-w-*. This occurred, however, only with dental finals. In cases like the following where we find alternation between labials and *l* in the same *hsieh-sheng* series with finals of other kinds we find no labial medial after *l-*:

里 l[?]: 埋 merj
 廬 l[?]: 膚 puā

This suggests that there must have been a rule to simplify *Cw-* clusters that arose by the processes we have described before finals other than dentals or *-j*. It is difficult to find clear evidence for this at present, however.

Conclusion

To sum up: my reconstruction of the *Shih-ching* rhyme categories listed on p. 181 is as follows:

	(a) <i>yin</i>	(b) <i>yang</i>	(c) <i>ju</i>
I		-əm	-əp
II		-am	-ap
III	-əl	-ən	-ət
IV(i)	-ats > -as	-an	-at
(ii)	-al > -aj		
V	-əj	-əŋ	-ək
VI	-aj > /ai/ [ɛ]	-aŋ	-ak
VII	-əɣ > /əɨ/ [ɻ]	-əŋ	-ək
VIII	-aɣ > /aɨ/ [a]	-aŋ	-ak
IX	-əw	-əŋ ^w	-ək ^w

X	-aw > /au/ [ɔ]	-aŋ ^W	-ak ^W
XI	-a ɰ > -aβ		-aq

The secondary values given after some of the *yin* rhymes are those which must be assumed for the Han Period.

Appendix

The Development of the *Ch'ieh-yün* Rhymes out of the *Shih-ching* Catagories

In the following schema I give the principal *Ch'ieh-yün* rhymes derived from each *Shih-ching* category, leaving out of consideration many special phenomena.

- I(b) A. óm > əm 覃 jám > em 添 rám > eɾm 咸
 B. òm > im 侵 wòm > uŋ 東 (e.g. 風 puŋ)
- (c) A. ép > əp 合 jáp > ep 帖 ráp > eɾp 洽
 B. èp > ip 緝
- II(b) A. ám > am 談 jám > em 添 rám > aɾm 銜 rjám > eɾm 咸
 B. àm > iǎm 嚴 jàm > iǎm 鹽 wàm > uǎm 凡
- (c) A. áp > ap 盍 jáp > ep 帖 ráp > aɾp 狎 rjáp > eɾp 洽
 B. àp > iǎp 業 jàp > iǎp 葉 wàp > uǎp 法
- III(a) A. ól > əj 治 jál > ej 齊 ról > eɾj 皆 wól > oj 灰
 B. òl > ij 微 jəl > ij 脂 wəl > uj 微
- (b) A. ón > ən 痕 jón > en 先 rón > eɾn 山 wón > on 魂
 B. òn > in 殷 jòn > in 眞 wòn > un 文
- (c) A. át > ət 沒 ját > et 屑 rát > ert 黠 wát > ot 沒
 B. èt > it 迄 jət > it 質 wət > ut 物
- IV(i)(a) A. áts > ajh 泰 játs > ejh 霽 ráts > aɾjh 夫 rjáts > eɾjh 皆
 B. àts > iǎjh 廢 jàts > iǎjh 祭 wàts > uǎjh 廢
- (b) A. án > an 寒 ján > en 先 rán > aɾn 刪 rján > eɾn 山
 B. àn > iǎn 元 jàn > iǎn 仙 wàn > uǎn 元
- (c) A. át > at 末 jat > et 屑 rát > aɾt 鎋 rját > eɾt 黠
 B. àt > iǎt 月 jət > iǎt 薛 wət > uǎt 月
- (ii)(a) A. ál > a 歌 rál > aɾ 麻
 B. àl > iǎ 支 à: l(?) > iǎɾ 麻
- V(a) A. ój > ej 齊 rój > eɾj 皆

- B. əj > ij 脂
- (b) A. əŋ > en 先 rəŋ > eŋ 山
B. əŋ > in 眞
- (c) A. əc > et 屑 rəc > eɾt 黠
B. əc > it 質
- VI(a) A. əj > ej 齊 rəj > eɾ(j) 佳
B. əj > (j)ǎ 支
- (b) A. əŋ > eŋ 青 rəŋ > eŋ 耕
B. əŋ > iɛŋ 清 rəŋ > iɛŋ 庚 (e.g. 敬 kiɛŋ^h)
- (c) A. əc > ek 錫 rəc > eɾk 麥
B. əc > iɛk 昔
- VII(a) A. əɣ > əj 昭 rəɣ > eɾj 皆 wəɣ > ow 侯 wə.ɣ(?) > oj 灰
B. əɣ > ɨ 之 wəɣ > uw 尤 rwəɣ > wij 脂
- (b) A. əŋ > əŋ 青 rəŋ > eɾŋ 耕 wəŋ > oŋ 東
B. əŋ > ɨŋ 蒸 wəŋ > uŋ 東
- (c) A. ək > ək 德 rək > eɾk 麥
B. ək > ɨk 職 wək > uk 屋
- VIII(a) A. əɣ > ɔ 模 rəɣ > aɾ 麻
B. əɣ > ɨa 魚 wəɣ > ua 虞 à:ɣ(?) > iaɾ 麻
- (b) A. əŋ > aŋ 唐 rəŋ > eɾŋ 庚
B. əŋ > ɨaŋ 陽 rəŋ > iɛŋ 庚
- (c) A. ək > ək 鐸 rək > eɾk 陌
B. ək > ɨak 藥 jək > iɛk 昔 rək > iɛɾk 陌
- IX(a) A. əw > aw 豪 rəw > aɾw 肴 jəw > ew 蕭
B. əw > uw 尤 jəw > iw 幽
- (b) A. əŋ^w > ɔŋ 冬 rəŋ^w > ɔŋ 江
B. əŋ^w > uŋ 東
- (c) A. ək^w > ɔk 沃 rək^w > ɔɾk 覺 jək^w > ek 錫
B. ək^w > uk 屋
- X(a) A. əw > ow 侯 rəw > (r) ow > (r) uw 尤 (e.g. 鄒 tɕuw)
B. əw > uǎ 虞
- (b) A. əŋ^w > oŋ 東 rəŋ^w > ɔŋ 江
B. əŋ^w > uɔŋ 鍾
- (c) A. ək^w > ok 屋 rək^w > ɔɾk 覺
B. ək^w > uɔk 燭
- XI(a) A. əɸ > aw 豪 rəɸ > aɾw 肴 jəɸ > ew
B. əɸ > ɨǎw
- (c) A. əq > ək 鐸 rəq > ɔɾk 覺 jəq > ek 錫 wəq > ɔk 沃 (~ok 屋)
aq > ɨak 藥

Bibliography

- Benedict, Paul K. 1972. *Sino-Tibetan: A conspectus*. Cambridge University Press.
- Bodman, N. C. 1954. *A linguistic study of the Shih-ming*. Cambridge, Mass.: Harvard University Press.
- Chou Fa-kao 周法高. 1969. "Lun shang-ku yin" 論上古音, *Journal of the Institute of Chinese Studies of the Chinese University of Hong Kong* 2/1, 109–178.
- Hashimoto, Mantarō, 1970. "Internal evidence for Ancient Chinese palatal endings," *Language* 46, 336–65.
- Haudricourt, A. 1954. "Comment reconstruire le chinois archaïque," *Word* 10, 351–64.
- Henderson, E. J. A. 1966. "Towards a prosodic statement of Vietnamese syllable structure," in C.E. Bazell, et al., eds., *In memory of J.R. Firth*. London: Longmans.
- Karlgren, B. 1957, *Grammata Serica recensa*, reprinted from *BMFEA* 29.
- Kōno Rokurō 河野六郎. 1968. *Chōsen kanjion no kenkyū* 朝鮮漢字音の研究, reprinted from *Chōsen gakuho* 31–35 and 41–44 (1964–67).
- Li Fang-kuei. 1945. "Some old Chinese loanwords in the Tai languages," *HJAS* 8, 333–342.
- . 李方桂 1972. "Shang-ku yin yen-chiu" 上古音研究, *Tsing-hua Journal of Chinese Studies* 9, 1–61.
- Lo Ch'ang-p'ei 羅常培 and Chou Tsu-mo 周祖謨. 1958. *Han Wei Chin Nan-pei ch'ao yün-pu yen-pien yen-chiu* 漢魏晉南北朝韻部演變研究 I, Peking: K'o-hsüeh ch'u-pan she.
- Mei Tsu-lin. 1970. "Tones and prosody in Middle Chinese and the origin of the rising tone," *HJAS* 30, 86–110.
- Pelliot, Paul. 1914. "Les noms propres dans les traductions chinoises du Milindapañha," *Journal Asiatique*, ser. 11, t.17, 317–419.
- . 1933. "Pāpīyān > Po-siun," *T'oung Pao* 30, 85–99.
- Pulleyblank, E. G., 1962. "The consonantal system of Old Chinese," *Asia Major* 9, 59–114 and 206–265.
- . 1963. "An interpretation of the vowel systems of Old Chinese and Written Burmese," *Asia Major* 10, 200–221.
- . 1965. "The transcription of Sanskrit *k* and *kh* in Chinese," *Asia Major* 11, 199–210.
- . 1965b. "Close/open ablaut in Sino-Tibetan," *Lingua* 14, 230–240.
- . 1969. "The semivowel *i* in Vietnamese and Mandarin Chinese," *BIHP* 39, 203–218.
- . 1970–71. "Late Middle Chinese," *Asia Major* 15/2, 197–239 and

- 16, 121–168.
- _____. 1972. "The analysis of vowel systems," *Acta Linguistica Hafniensia* 14, 39–62.
- _____. 1973a. "Some new hypotheses concerning word families in Chinese," *Journal of Chinese Linguistics* 1, 111–125.
- _____. 1973b. "Some further evidence regarding Old Chinese -s and its time of disappearance," *BSOAS* 36, 368–73.
- _____. 1979. "Linguistic evidence for the date of Han-shan," in Ronald C. Miao, ed., *Studies in Chinese Poetry and Poetics*, vol. 1, pp. 163–195. San Francisco: Chinese Material Center.
- Rai Tsutomu 頼惟勤. 1953. "Jōko Chūgoku [sic] go no kōon imbi ni tsuite" 上古中國 [sic] 語の喉音韻尾について, *Ochanomizu joshi daigaku jimbun kagaku kiyō* お茶の水女子大學人文科學紀要 3, 51–64.
- _____. 1957. "Jōko Chūgokugo no imbo ni kansuru nisan no mondai" 上古中國語の韻母に関する二三の問題, *Tōyō gakuho* 40–41, 62–81.
- Schuessler, A. 1974. "Final -l in Archaic Chinese," *Journal of Chinese Linguistics* 2, 27–36.
- Tung T'ung-ho 董同龢. 1944. "Shang-ku yin-yün piao-kao" 上古音韻表稿, *BIHP* 18, 1–249.
- Wang Li 王力. 1958. *Han yü shih kao* 漢語史稿, 3 vols. Peking: K'o-hsüeh ch'u-pan she.
- Yakhontov, S. 1960. "Fonetika kitaiskogo jazyka I tysjačletija do i. e.," *Problemy vostokovedeniya* 6, 102–115. Translated into English by J. Norman, "The phonology of Chinese of the first millennium B.C. (Rounded vowels)," *Unicorn* 6 (1970) 52–75.